

Lecture 8 Pf of Sz.R.L.

- i) RL : Ramsey-Turán upper bd $\rightsquigarrow$ weighted Turán prob.
- ii) bd deg H has linear Ramsey #.
- iii) Spectral pf of RL.

$$RT(n, K_4, o(n)) = \max \# \text{ edges in an } n\text{-vc } K_4\text{-free graph } G \text{ with } \alpha(G) = o(n)$$

Thm (Sz 76) $RT(n, K_4, o(n)) \leq \frac{n^2}{8} + o(n^2)$

Pf: • Let G be as above. $0 < \epsilon < \delta < 1$

$G \xrightarrow{RL} R$ reduced graph $R(\epsilon, \delta)$

Claim Suffices to show $\begin{cases} 1) R \text{ is } \Delta\text{-free} \\ 2) R \text{ has no chubby edge} \end{cases}$
 $i, j \in E(R), d(v_i, v_j) \geq \frac{1}{2} + o(1)$

If true, $2) \Rightarrow \forall i, j \in E(R) \Rightarrow d(v_i, v_j) \leq \frac{1}{2} + o(1)$

$\begin{matrix} \gamma_i & \gamma_j \\ \bigcirc & \bigcirc \\ \leq \frac{1}{2} + o(1) \end{matrix}$

$1) \Rightarrow$ Turán Thm $e(R) \leq \frac{r^2}{4}$

$$e(G) \leq e(R) \cdot \frac{n^2}{r^2} \left(\frac{1}{2} + o(1) \right) + (e(G) - e(G_R))$$

$$= \frac{r^2}{4} \cdot \frac{n^2}{r^2} \left(\frac{1}{2} + o(1) \right) + o(n^2)$$

$$= \frac{n^2}{8} + o(n^2)$$

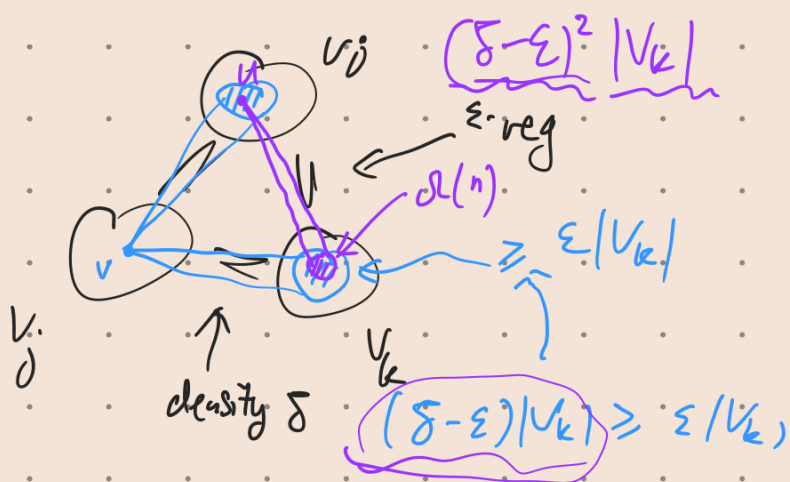
$\left. \begin{matrix} \text{edges cleaned} \\ \text{is sparse pairs} \\ \text{irreg pairs} \\ v_i, i \in [r] \end{matrix} \right\} o(n^2)$

• Suppose $\Delta \in R$ i, j, k

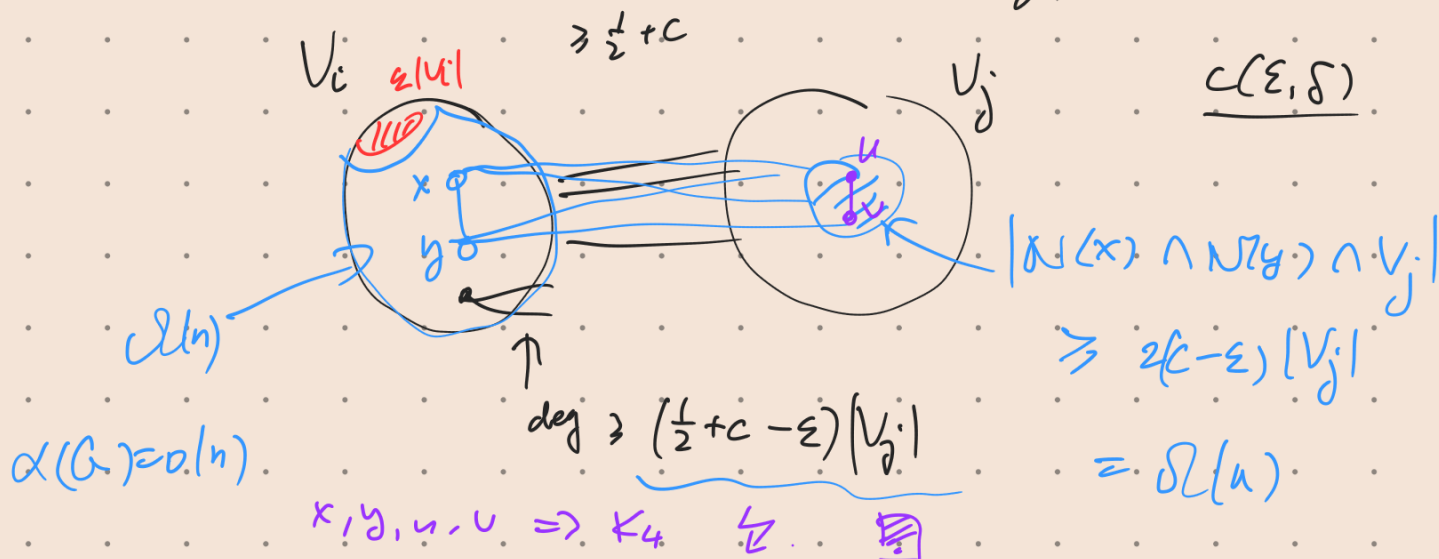
$$\Rightarrow \text{in } G$$

$$\chi(G) = \chi(n)$$

$\Rightarrow$ purple point has
an edge $+ u, v$
 $\Rightarrow K_4$



• Suppose $\exists i, j \in E(R)$ s.t. $d(V_i, V_j) \geq \frac{1}{2} + c$



Chvátal - Rödl - Szemerédi - Trotter - Thm

Def. G graph. $\Gamma(G, G) = \min N$ s.t.

$\forall$ 2-edge-colouring of $K_N \Rightarrow \exists$ monochromatic G .

- $G = K_n$, $2^{n/2} \leq r(K_n, K_n) < 4^n$
exponential in $|G|$

$$\left[\begin{array}{l} \text{Thm (CRS}_{\mathbb{Z}T}) \quad \forall d \in \mathbb{N}, \quad \forall G \text{ w. } \Delta(G) \leq d \\ \Rightarrow \quad r(G, G) = O_d(n) \quad \left(r(G, G) \leq C(d) \cdot n \right) \end{array} \right]$$

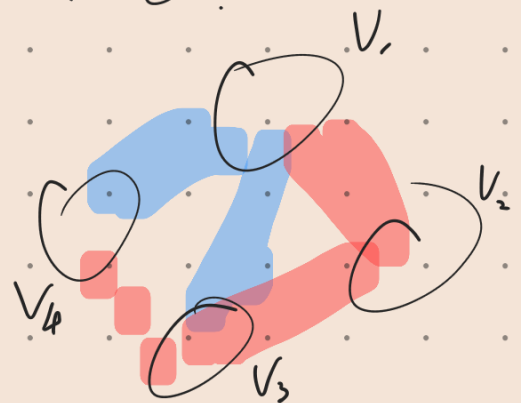
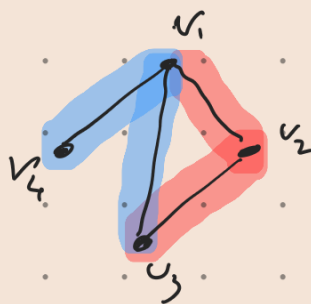
Recall Thm Brook's Thm: $\forall H$ w./ $\Delta(H) \leq d$, $\chi(H) \leq d+1$.

($k=2$)

- For a k -edge-coloured graph G , a partition $V(G) = V_1 \cup \dots \cup V_r$ is ε -regular if
 - $||V_i| - |V_j|| \leq 1$
 - for all but $\leq \varepsilon \binom{r}{2}$ pairs $i, j \in \left[\frac{r}{2} \right]$, (V_i, V_j) is ε -regular in every colour.

Lem (Multi-col R.L.) $\forall \varepsilon > 0, \forall k \in \mathbb{N}, \forall m, \exists M = M(\varepsilon, k, m)$ s.t. $\forall k$ -edge-col. G with $n \geq m$ v.s admits an ε -reg partition $V_1 \cup \dots \cup V_r, m \leq r \leq M$.

Given a reg partition of edge-col G .



Pf (CRS₂T)

$$r(G, G) = O_d(n)$$

$$\begin{cases} \Delta(G) = d \\ |G| = n \end{cases}$$

upper bound on # parts

$$\text{Take } C = \frac{2M}{(\frac{1}{2} - \varepsilon)^d}$$

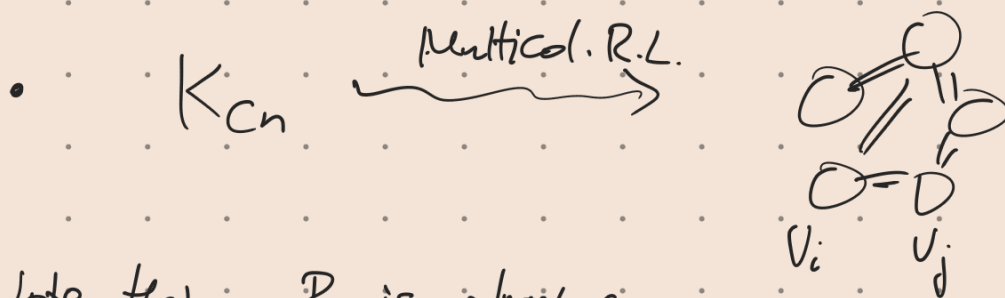
, where $M = M(\varepsilon, m, 2)$ in MRL

$$r(G, G) \leq C \cdot n$$

$$m \geq 5r(K_{d+1}, K_{d+1})$$

Fix an arbitrary 2-edge-col. of K_{cn}

NT embed monoX copy of G in K_{cn} .



blue edges $\geq \frac{1}{2}$ proportion

Note that R is almost complete

$$e(R) \geq (1-\varepsilon) \binom{n}{2} = \left(1 - \frac{1}{m}\right) \binom{n}{2}$$

by choosing $m = \frac{1}{\varepsilon}$, $\varepsilon = \frac{1}{m}$

$$\frac{n}{3} > r(K_{d_1}, K_{d_2})$$

by Turán thm $\Rightarrow R \geq K_{n/3}$

$\Rightarrow \exists \text{ monoX. } K_{d_1} \text{ in } R$

$$\Delta(G) \leq d$$

$\exists \text{ homo.}$

$$G \rightarrow K_{d_1}$$

$$\begin{array}{ccc} \uparrow & \text{embedding} & \uparrow \\ K_{cn} & \xleftarrow{\text{Lem}} & R \end{array}$$



Lem G n -vx, $\varepsilon > 0$. Then there exists

a partition $V = V_1 \cup \dots \cup V_M$, $M \leq M(\varepsilon)$

s.t. apart from an exceptional set

$$\Lambda \subseteq \binom{[M]}{2} \text{ with}$$

$$\sum_{i,j \in \Lambda} |V_i||V_j| = O(\varepsilon |V|^2)$$

we have $\forall i,j \notin \Lambda$, $A \in V_i$, $B \in V_j$ that

$$|e(A,B) - d_{ij} |A||B|| = O(\varepsilon |V_i||V_j|)$$

$$e(A,B) = \mathbf{1}_A^* T \mathbf{1}_B = \mathbf{1}_A^* T_1 \mathbf{1}_B + \mathbf{1}_A^* T_2 \mathbf{1}_B + \mathbf{1}_A^* T_3 \mathbf{1}_B$$

Idea $T = \text{adj matrix of } G \text{ sym. real}$

$$= \sum_{i=1}^n \lambda_i u_i u_i^* \quad (\text{eigenvalue decomp.})$$

partition T into main term (λ_i large)

error term (λ_i small)

Will partition $V(G)$ s.t. every 'heavy' eigenvector does NOT fluctuate much in each part.

Pf: $T = \sum_{i=1}^n \lambda_i u_i u_i^* \quad |\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$

Split T $T = T_1 + T_2 + T_3$

$T_{ij}^k = \# \text{ length } k\text{-walk from } u_i \text{ to } u_j \text{ in } G$

$$\sum_{i=1}^n |h_i|^2 = \text{tr}(T^2) = \sum d_i = 2e(G) \leq n^2$$

(\) $\sum |\lambda_i|^2 \leq n^2 \Rightarrow |\lambda_i| \leq \frac{n}{\sqrt{c}}$

Take $F = F(\varepsilon) : N \rightarrow N$ w/ $F(i) \geq c$

By averaging, $\exists J \leq F^{\frac{1}{\varepsilon^3}}(1)$ s.t. $F^2(1) = F(F(1))$

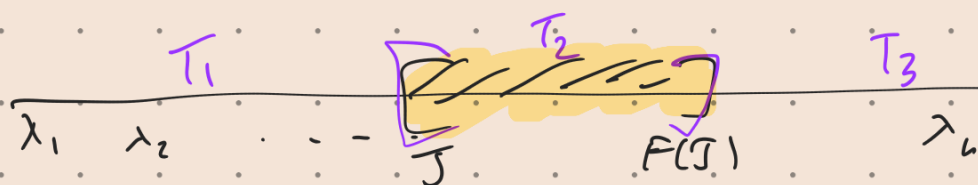
$$\sum_{i \in [J, F(J)]} |\lambda_i|^2 \leq \varepsilon^3 n^2$$

$$\underbrace{F \circ F \circ \dots \circ F(1)}_{1/\varepsilon^3}$$

$$T_1 = \sum_{i \in J} \lambda_i u_i u_i^*$$

$$T_2 = \sum_{J \leq i \leq F(J)} \lambda_i u_i u_i^*$$

$$T_3 = \sum_{i \geq F(J)} \lambda_i u_i u_i^*$$



(i) Partition for the structured part T_1 .

(ii) bad error terms T_2 & T_3 .

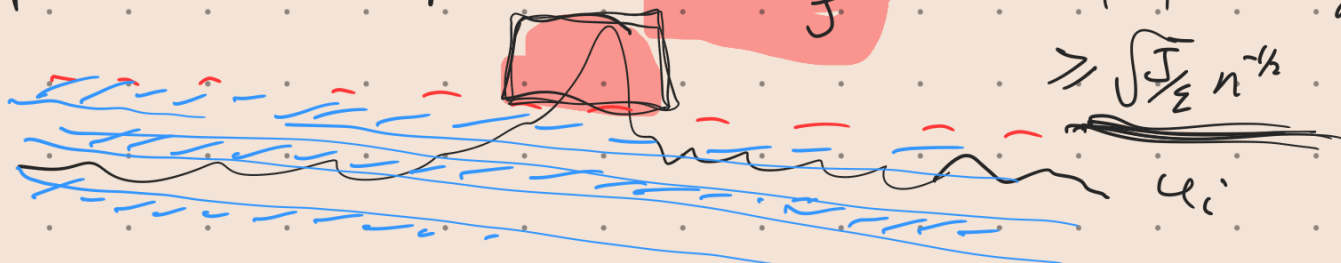
For (i) Want to construct a partition of $V(G)$ s.t.

$T_1 = \sum_{i \in J} \lambda_i u_i u_i^*$ is approximately constant on most parts.

• For each $i \in J$, partition $V(G)$ into $O_{J, \varepsilon}(1)$ parts

in which u_i only fluctuates by $O\left(\frac{\varepsilon^{3/2}}{J} n^{-1/2}\right)$

apart from an exceptional set $\leq \frac{\varepsilon n}{J}$ where $|u_i|$ too large



Take $u = u_i$, $u(j) = j^{\text{th}}$ coord. of u

$$\|u\|_2^2 = \sum_{j \in [n]} u_j^2 = 1.$$

large coord. $\leq \frac{\varepsilon n}{J}$

$$\frac{\varepsilon n}{J} \cdot \left(\sqrt{\frac{J}{\varepsilon}} n^{-1/2}\right)^2 > 1$$

For the rest of the coordinates part. it into

$$\frac{\sqrt{\frac{J}{\varepsilon}} n^{-1/2}}{\frac{\varepsilon^{3/2}}{J} n^{-1/2}} = \# \text{ parts} = O_{J, \varepsilon}(1)$$

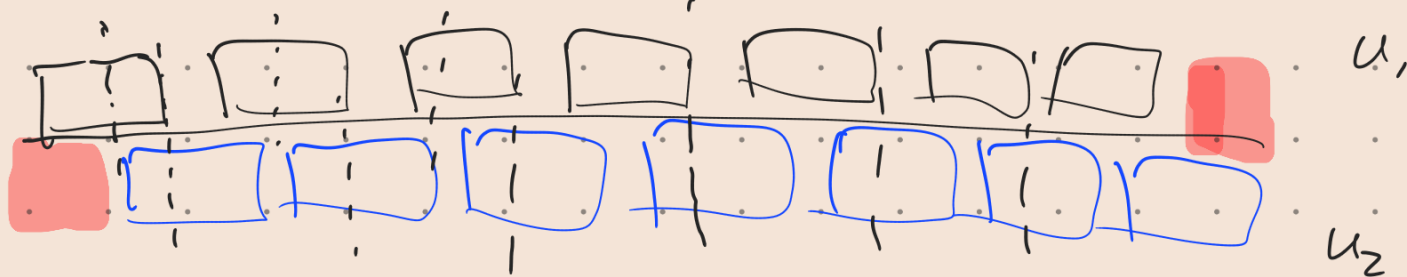
↑
fluctuation

Combine all J partition together

$$\Rightarrow V(G) = V_1 \cup \dots \cup V_{M-1} \cup V_M$$



$$u_i = u$$



$$\mathbb{1}_A^* T_1 \mathbb{1}_B = \text{dig } |A||B| + O(\varepsilon |V_i| |V_j|) \quad T_1 = \sum_{i \in J} \lambda_i u_i u_i^*$$

$$|\lambda_i| \leq \frac{n}{\sqrt{j}} \quad (*)$$

v_i, v_j -entry of $T_1 = \sum \lambda_i u_i u_i^*$

fluctuates at most

$$\sum_{i \in J} \lambda_i \underbrace{\sqrt{\frac{j}{\varepsilon}} n^{-1/2}}_{\text{max value}} \cdot \underbrace{O\left(\frac{\varepsilon^{3/2}}{j} n^{-1/2}\right)}_{\text{fluctuation}} \stackrel{(*)}{\leq} O(\varepsilon)$$



$$\sum \lambda_i \overset{\text{max}}{u_i} u_i^* \stackrel{\text{dig}}{=} \text{dig}$$

Left to show
 $\forall k \in \{2, 3\}$

$$\mathbb{1}_A^* T_k \mathbb{1}_B = O(\varepsilon |V_i| |V_j|)$$

$$e(A, B) = \sum_{k \in [3]} \mathbb{1}_A^* T_k \mathbb{1}_B = \text{dig } |A||B| + O(\varepsilon |V_i| |V_j|)$$

$$\mathbb{1}_A^* T_2 \mathbb{1}_B = O(\varepsilon |V_i| |V_j|)$$

$$\text{b/c} \quad \sum_{i \in (J, F(J))} |\lambda_i|^2 \leq \varepsilon^3 n^2$$

$$\mathbb{1}_A^* T_3 \mathbb{1}_B \leq O\left(\frac{n^2}{\sqrt{F(J)}}\right) \quad F: N \rightarrow N$$

$$= O(\varepsilon |v_i| |y_j|).$$